

Machine Learning Engineer with ~3 years of experience specializing in Generative AI, multi-agent orchestration, MLOps, and enterprise cloud architecture on GCP. Proven track record of architecting scalable retrieval systems (RAG), and deploying microservices-backed production AI platforms for enterprise data analytics.

EXPERIENCE

Machine Learning Engineer <i>Quantiphi</i> <i>RCA, Segmentation & NL2SQL Bot</i>	Bengaluru, IN <i>Feb 2024 – Present</i> <i>AI Agents, ADK, BigQuery, Apache Kafka, RAG, Multi Agent Orchrestration</i>
- Led a team of 5+ ML engineers within a 24+ member engineering group to design and ship 3 production-grade multi-agent orchestration systems, enabling executive teams to query enterprise metrics via natural language. - Architected an agentic analytics platform utilizing Google Agent Development Kit (ADK) with a custom RAG retrieval engine indexing schemas across 14+ structured data sources (Kafka topics, Google Ads, BigQuery) for precise NL2SQL synthesis. - Cut executive & Managements data-reporting turnaround time from days to under 2 minutes, accelerating decision-making velocity across business units.	
<i>Conversational Search & Recommendation Engine</i>	<i>Vertex AI Search, Dynamic Model Swap, Ranking, Keyword Extraction</i>
- Architected a real-time conversational search and recommendation engine, replacing legacy fuzzy keyword matching with intent-aware semantic vector retrieval. - Engineered a high-throughput search platform using Vertex AI Search for Commerce, indexing an 1.5M+ SKU product catalog and implementing custom ranking models (frequently-bought-together, similar items, personalized scoring). - Increased catalog discovery accuracy and user engagement by aligning free-form natural language queries directly to product vector spaces.	
<i>Multimodal Email Content Generation</i>	<i>Agentic Workflows, Imagen, Cloud Run, Web Scraping</i>
- Built an automated multimodal marketing workflow that extracts brand aesthetics and metadata from target web domains to generate custom email assets. - Engineered headless scraping pipelines (Playwright) to extract design assets, feeding structured contextual prompts into Gemini and Imagen models deployed on Cloud Run for dynamic asset generation. - Reduced marketing campaign creation cycles from weeks to under 5 minutes by completely automating asset extraction and copy generation.	
<i>Lead Hotspot Predection</i>	<i>Hybrid Predective Model, SDWIS Database, Geospatial Feature Engineering</i>
- Engineered machine learning pipelines to predict municipal lead contamination risks across regional drinking water systems. - Preprocessed and engineered spatial features from Public Water System (PWS) datasets sourced from the EPA SDWIS database, implementing hybrid XGBoost and heuristic rule-based decision trees. - Achieved 88+% predictive accuracy for lead hotspot identification, directly optimizing resource allocation for infrastructure remediation teams.	
Software Engineer Intern	Remote
<i>JP Morgan Chase & Co.</i>	<i>Nov 2022 – Sept 2023</i>
- Engineered a real-time financial streaming interface capable of ingesting and processing over 100,000 data points/second with minimal latency. - Utilized JP Morgan’s Perspective open-source data engine to construct dynamic visualization dashboards monitoring historical stock pair correlations. - Implemented automated bound-crossing alert triggers based on statistical correlation metrics to support real-time quantitative monitoring.	

EDUCATION

B.M.S. College of Engineering <i>Bachelor of Engineering – Electronics & Instrumentation Engineering; GPA: 8.3/10</i>	Bangalore, India <i>2020 – 2024</i>
S.Tech IT School <i>Higher Secondary School Certificate – PCM; Percentage: 82.6%</i>	Rajasthan, India <i>2019 – 2020</i>

TECHNICAL SKILLS

Programming & Core CS: Python, C/C++, Java, SQL, Bash / Shell Scripting, Object-Oriented Programming (OOP)
Generative AI & Agentic Systems: Google Agent Development Kit (ADK), LLMs, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LangChain, LangGraph, Prompt Engineering, Vector Search, Hybrid Search, Embeddings, Fine-Tuning, PEFT/LoRA, AI Guardrails, Nvidia NeMo
Machine Learning & Data Science: Neural Networks, PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, Pandas, NumPy, OpenCV, Natural Language Processing (NLP), Computer Vision, Feature Engineering, Exploratory Data Analysis (EDA), Time-Series Forecasting
GCP, Cloud & Infrastructure: Google Cloud Platform (GCP), Vertex AI, Search/Recommendation Models, Cloud Run/AWS Lambda, Cloud Functions, Google Cloud Storage (GCS), Firebase, Pub/Sub, AWS, Azure, Docker, Kubernetes, DevOps, Linux, Cloud Technologies
Data Engineering & Architecture: BigQuery, Data Warehousing, Apache Kafka, Snowflake, PostgreSQL, MongoDB, MySQL, Redis, NoSQL, Vector Databases, Data Pipelines, Microservices, REST APIs, FastAPI, Backend Architecture, Full-Stack Integration, System Design
MLOps, Web & Tools: CI/CD, MLflow, Hugging Face, Model Deployment, Streamlit, Git, GitHub, Document AI, Agile Methodologies

CERTIFICATIONS AND ACHIEVEMENTS

Associate Cloud Engineer, Google Cloud
One for All, Quantiphi
Think Tank, Quantiphi
Unsung hero - Consider It Done, Quantiphi